

**Department of Electrical and Electronics Engineering****Academic Year 2024 – 2025 (Even Semester)****Degree, Semester & Branch:** IV Semester B.E. EEE**Course Code & Title:** EE3401 Transmission and Distribution**Name of the Faculty member (s):** Dr. S. Narthana**Innovative Practice Description**

- **Unit / Topic:** Unit I / Types of transmission line conductors
- **Course Outcome:** CO 1
- **Unit Outcome:** TLO2
- **Activity Chosen:** One Minute Paper
- **Justification:**

The chosen topic - types of conductors have 3 types in transmission and distribution conductors. After teaching the concept, I thought of conducting this activity for enabling the students to give the difference between each type of conductors which enhance their learning level and as a teacher I can judge the understanding level of the students.

- **Time Allotted for the Activity:** 10 minutes

- **Details of the Implementation:**

At the end of the class,

- ✓ I asked the students to think about various types of conductors in transmission and distribution lines for 5 minutes.
- ✓ I told them to write as much as they remember for 3 minutes.
- ✓ Finally, I collected the papers from the students (2 minutes).

This activity shows whether the students can able to understand the specific topic and their involvement in the particular class.

- **CO – PO / PSO mapping:**

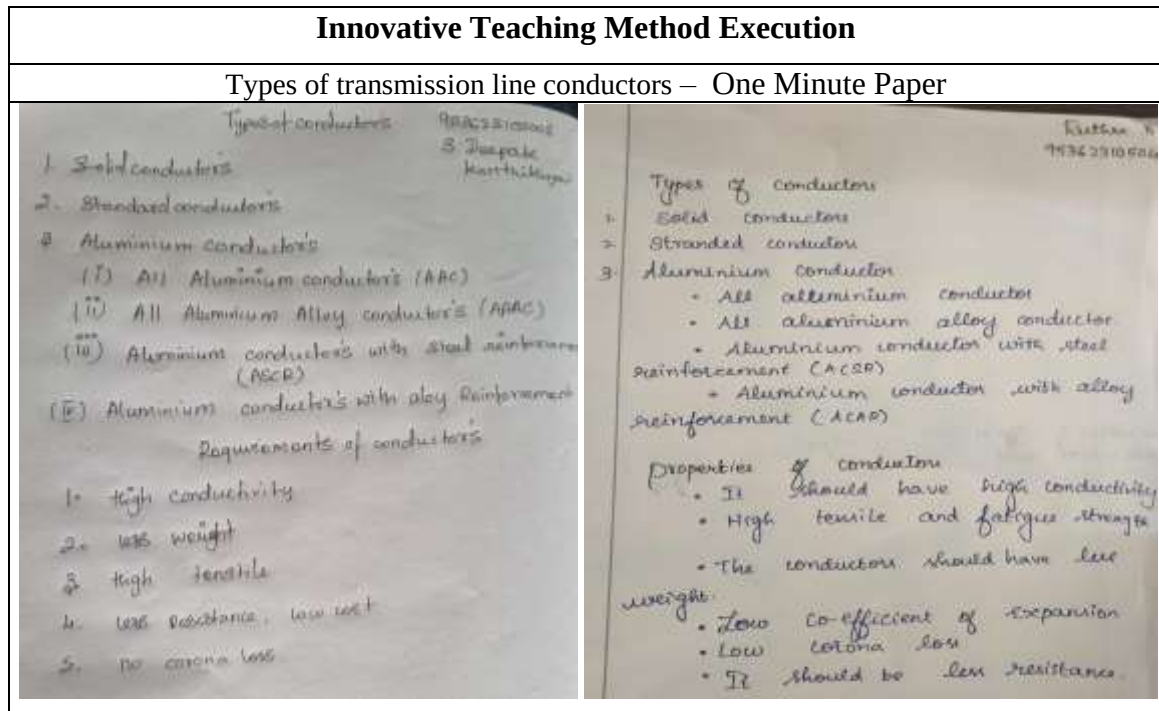
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	1	-	-	-	-	1	2	1	-	1	2	-	-

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO 9
	2
Justification for correlation	The students can function effectively as an individual

- Images / Screenshot of the practice:



Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

❖ *Benefit of the practice:*

- ❖ All the students were able to write the given topic.
- ❖ After conducting the activity, I came to know that students understood the topic and they were able to explain it.

❖ *Challenges faced in implementation:*

1. Time utilization for conducting the activity
2. Some students could not be able to recollect and write immediately.

References:

1. D.P.Kothari, I.J. Nagarath, 'Power System Engineering', Mc Graw-Hill Publishing Company limited, New Delhi, Third Edition, 2019.
2. V.K.Mehta, Rohit Mehta, 'Principles of power system', S. Chand & Company Ltd, New Delhi, 2013
3. <https://oncourseworkshop.com/self-awareness/one-minute-paper/>

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Department of Electrical and Electronics Engineering Academic Year 2024 – 2025 (Even Semester)

Degree, Semester & Branch: IV Semester B.E. EEE

Course Code & Title: EE3401 Transmission and Distribution

Name of the Faculty member (s): Dr. S. Narthana

Innovative Practice Description

- **Unit / Topic:** Unit II / Corona
- **Course Outcome:** CO 2
- **Unit Outcome:** TLO6
- **Activity Chosen:** Think Pair Share
- **Justification:**

The chosen topic – corona provides safety to the transmission lines and reduces flash over. After teaching the concept, I thought of conducting this activity for making the students to provide knowledge about corona and its effects which enhance the learning level and as a teacher I can judge the understanding level of the students.

Time Allotted for the Activity: 10 minutes

- **Details of the Implementation:**

After completing the topic, the students will be paired with their neighbors, 3 students as a group.

At the end the Class (Last 10 minutes)

- I asked the students to **think** about corona, factors affecting corona and its loss for 2 minutes. Then I told them to **Pair** with their neighbors and discuss about the concept of corona in transmission line for another 2 minutes.
- Finally, I selected 1 Pair from each column randomly and ask them to **share** about corona and its advantages. (6 minutes)
- Finally, I summarized the points again about corona in transmission line.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO2	3	2	1	-	-	-	-	1	2	1	-	1	2	-	-

(1 – Low 2 – Moderate 3 – High)

- PO / PSO mapped:

Innovative practice	PO 9
	1
Justification for correlation	Due to this activity, student's team work is getting enhanced. So it is slightly correlated.

- Images / Screenshot of the practice:

Innovative Teaching Method Execution

Corona – Think Pair Share

Corona

The Overhead lines there exists a hissing noise and sometimes a faint violet glow, the effect due to which such phenomenon exists surrounding the overhead lines, is called corona effect. The phenomenon of hissing noise, faint violet glow and production of ozone gas surrounding the overhead lines, due to ionization of air is called corona.

Methods of Reducing corona effect

The corona effects are observed when the operating voltage level is more than 33KV.

* Increasing the conductor size:

By increasing the size of the conductor disruptive voltage is raised

* Increasing the conductor spacing:

The increased spacing increases the voltage level at which corona starts. Thus the corona effects are reduced.

* Using hollow and bundled conductors:

The hollow and bundled conductor reduce the corona effects

Factors affecting Corona and Corona loss

* Electrical factors:

Higher the supply frequency, higher is the corona loss. Thus d.c. corona loss is less compared to a.c. corona loss.

* Line voltage:

The line voltage directly affects the corona and corona loss. For lower line voltage corona may be absent

* Surface conditions:

The corona loss is directly proportional of square root of radius of conductor. The corona depends on the surface conditions.

* Spacing between conductor:

If the spacing is made very large, corona can be absent, particularly the spacing is selected so that corona tolerable.

* Shape of conductors:

The shape of the conductor from the ground also like flat, oval, cylindrical etc. affects the corona loss.

Advantages:

The air surrounding the conductor is ionised and become conducting. This increases the virtual diameter of the corona

Disadvantages:

The corona power loss is the biggest disadvantage which reduce the transmission efficiency.



Vadakku Venganallur, Tamil Nadu, India
 Faj7+ww, Ramco Institute Of Technology Rd, Vadakku Venganallur, Tamil Nadu 626126, India
 Lat 9.482808° Long 77.514406°
 24/03/2025 11:21 AM GMT +05:30

Reflective Critique:

❖ ***Feedback of practice from students and other stakeholders:***

Students felt that discussing with the team members and sharing the concept with everyone make them understand it well and enhanced their knowledge.

❖ ***Benefit of the practice:***

1. All the students were able to share the topic given in an effective way.
2. Students understood the concept which was reflected from their answers during sharing session.

❖ ***Challenges faced in implementation:***

1. Time utilization for conducting the activity.
2. Slow learners were not able to recollect and present some points during sharing session.

References:

1. D.P.Kothari, I.J. Nagarath, 'Power System Engineering', Mc Graw-Hill Publishing Company limited, New Delhi, Third Edition, 2019.
2. V.K.Mehta, Rohit Mehta, 'Principles of power system', S. Chand & Company Ltd, New Delhi, 2013
3. <https://www.readingrockets.org/strategies/think-pair-share>

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- **Unit / Topic:** Unit III / Types of Towers

- **Course Outcome:** CO 3

- **Unit Outcome:** TLO7

- **Activity Chosen:** Ungraded Quiz

- **Justification:**

The concept and design aspects of the line supports is explained by this active learning approach. This activity provides a conceptual bridge between successive class periods. Improve the quality of student participation by recalling the important aspects about the involved in the design of various types of towers used for overhead transmission lines.

- **Time Allotted for the Activity:** 15 minutes

- **Details of the Implementation:**

Quiz activities require students to actively engage in their learning, often by connecting their prior knowledge to new information. When creating a quiz, a student frequently interacts with a textbook, notes from class, an instructor, classmate, or study group.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO3	3	2	2	3	-	-	-	1	2	1	-	1	2	-	-

(1 – Low 2 – Moderate 3 – High)

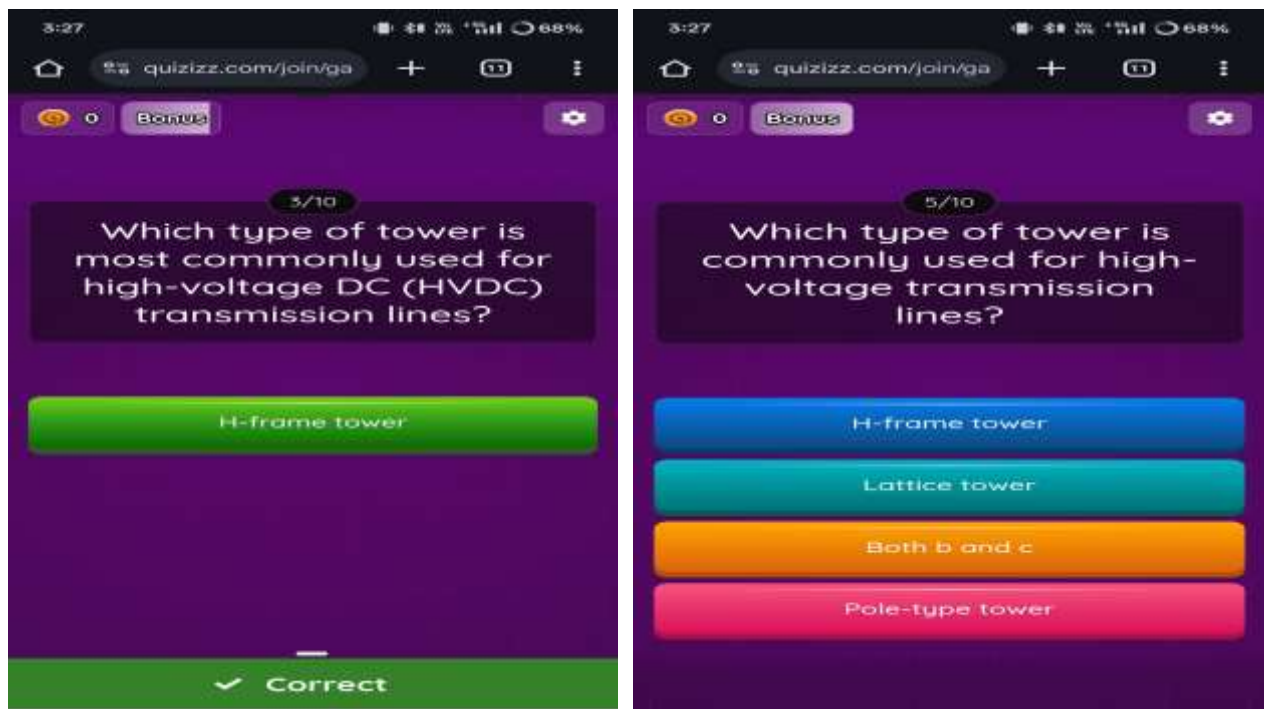
- **PO / PSO mapped:**

Innovative practice	PO 9
	1
Justification for correlation	Due to this activity, The students can function effectively as an individual.

- **Images / Screenshot of the practice:**

Innovative Teaching Method Execution

Types of Towers - Ungraded Quiz



Reflective Critique:

❖ **Feedback of practice from students and other stakeholders:**

Students understood the important points in design present in this topic and improved the ability to relate their answers to the classroom learning.

❖ **Benefit of the practice:** The benefits of quiz are a great way for students to make notes on all of the information they receive. It made the students to understand the various types of line supports.

❖ **Challenges faced in implementation:**

Normally teachers will give longer explanations in the notes section of the topic. The students are made to answer the multiple-choice questions, fill ups for the topic of towers only for 10 mins. But in real scenario, it takes 20 mins.

References:

- ❖ D.P.Kothari, I.J. Nagarath, 'Power System Engineering', Mc Graw-Hill Publishing Company limited, New Delhi, Second Edition, 2008.
- ❖ V.K.Mehta, Rohit Mehta, 'Principles of power system', S. Chand & Company Ltd, New Delhi, 2013

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- **Unit / Topic:** Unit III / Insulators and its Types
- **Course Outcome:** CO 3
- **Unit Outcome:** TLO10
- **Activity Chosen:** Demonstration
- **Justification:**

After teaching the concept, I thought of conducting this activity for making the students to give the exposure about the different types of Insulators and their application areas which enhance the learning level and as a teacher I can judge the understanding level of the students.

Time Allotted for the Activity: 15 minutes

- **Details of the Implementation:**

After completing the topic, three minutes is given to the students to think about the topic without writing anything.

At the end of the class

- ✓ I asked the students to think about constructional features and different types of Insulators for 2 minutes.
- ✓ Then I told them to Pair with their neighbors and discuss about the different types of Insulators for another 2 minutes.
- ✓ Then, I shown the demo Insulators for each student and explained. (8 minutes)
- ✓ Finally, I asked each student to tell about the application area. (3 Minutes)

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO3	3	2	2	3	-	-	-	1	2	1	-	1	2	-	-

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO 4
	3
Justification for correlation	Due to this activity, students will understand the environmental effects on conductor. So it is highly correlated.

- Images / Screenshot of the practice:



Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

Students were interested involving in the activity and requested to conduct more such activities in future.

❖ *Benefit of the practice:*

- ✓ The assessment of effectiveness of the activity was good in a way that while asking questions, most of the students answered and shared many points.
- ✓ After conducting the activity, I understood that the students could be able to recollect the points about the types of insulators and their application area.

❖ *Challenges faced in implementation:*

- ✓ Time is not enough for this activity, so will allot more time in future.

References:

- ❖ D.P.Kothari, I.J. Nagarath, 'Power System Engineering', Mc Graw-Hill Publishing Company limited, New Delhi, Second Edition, 2008.
- ❖ V.K.Mehta, Rohit Mehta, 'Principles of power system', S. Chand & Company Ltd, New Delhi, 2013
- ❖ <https://blog.teachmint.com/demonstration-method-an-introduction/#:~:text=In%20the%20demonstration%20method%20of,knowledge%2C%20thereby%20further%20enhancing%20it.>

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Department of Electrical and Electronics Engineering

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Course Code & Title: EE3401 Transmission and Distribution

Name of the Faculty member (s): Dr. S. Narthana

Innovative Practice Description

- **Unit / Topic:** Unit IV / Construction of Underground Cables
- **Course Outcome:** CO 4
- **Unit Outcome:** TLO11
- **Activity Chosen:** Demonstration
- **Justification:**

After teaching the concept, I thought of conducting this activity for making the students to give exposure about the constructional features and different insulating materials used in UG cables which enhance the learning level and as a teacher I can judge the understanding level of the students.

Time Allotted for the Activity: 20 minutes

- **Details of the Implementation:**

After completing the topic, three minutes is given to the students to think about the topic without writing anything.

At the end of the class

- I asked the students to think about constructional features and different types of Cables for 3 minutes.
- Then I told them to Pair with their neighbor and discuss about the different types of Cables and constructional features for another 3 minutes.
- Then, I shown the demo Cables for each student and explained. (10 minutes)
- Finally, I asked some students to explain about the types and constructional features. (4 minutes)

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO4	3	2	1	-	-	-	-	1	2	1	-	1	2	-	-

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO 7
	1
Justification for correlation	Due to this activity, students will understand the impact of professional engineering solution for environment. So it is slightly correlated.

- **Images / Screenshot of the practice:**



Reflective Critique:

- ❖ **Feedback of practice from students and other stakeholders:**

Students were interested involving in the activity and requested to conduct more such activities in future.

- ❖ **Benefit of the practice:**

- ✓ The assessment of effectiveness of the activity was good in a way that while asking questions, most of the students answered and shared many points.
- ✓ After conducting the activity, I understood that the students could be able to recollect the points about construction of cables and able to differentiate single core and three core cables.

- ❖ **Challenges faced in implementation:**

- ✓ Time is not enough for this activity, so will allot more time in future.

References:

- ❖ D.P.Kothari, I.J. Nagarath, 'Power System Engineering', Mc Graw-Hill Publishing Company limited, New Delhi, Second Edition, 2008.
- ❖ V.K.Mehta, Rohit Mehta, 'Principles of power system', S. Chand & Company Ltd, New Delhi, 2013
- ❖ <https://blog.teachmint.com/demonstration-method-an-introduction/#:~:text=In%20the%20demonstration%20method%20of,knowledge%2C%20thereby%20further%20enhancing%20it.>

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- **Unit / Topic:** Unit V / Types of Substations
- **Course Outcome:** CO 5
- **Unit Outcome:** TLO17
- **Activity Chosen:** Field Visit
- **Justification:**

The topic Substation Layout has different types which is used for various applications. After completing the topic, I took the students for field visit to enable the students to acquire more knowledge in observing the layout, isolator switches, various types of circuit breakers and transformers etc.

Time Allotted for the Activity: 45 Minutes

- **Details of the Implementation:**

After completing the topic, I took the students to the RML 110 kV substation and shown them how the input power and power distribution is carried out inside the unit. At the end of the class

- I requested all the students to discuss among themselves for 5 minutes.
- Finally, I asked questions to students about the substation layout and method of power distribution and the students answered well which shows the understanding of the topic through this field visit. (10 minutes)

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO5	3	2	2	3	-	2	-	1	2	1	-	1	2	-	-

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO 6
	2
Justification for correlation	Due to this activity, students will learn about the importance of substations and its impact on societal health and safety, so it is moderately correlated (level 2).

- Images / Screenshot of the practice:



Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

Students were interested in attending field visit and requested to conduct more such activities in future.

❖ *Benefit of the practice:*

- ✓ The assessment of effectiveness of the activity was good in a way that while asking questions, most of the students answered and shared many points.
- ✓ After conducting the activity, I understood that the students could be able to recollect the points about substation.

❖ *Challenges faced in implementation:*

- ✓ Time is not enough for this activity, so will allot more time in future.

References:

- ❖ D.P.Kothari, I.J. Nagarath, 'Power System Engineering', Mc Graw-Hill Publishing Company limited, New Delhi, Second Edition, 2008.
- ❖ V.K.Mehta, Rohit Mehta, 'Principles of power system', S. Chand & Company Ltd, New Delhi, 2013

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